

# COS 433: Cryptography

## Reminder:

- Problem Set 1 due 9/24.

# The Subset Sum Generator

$$G_A(\boxed{s}) = sA \pmod{2}$$

Poly-time attack:

- Given  $y \in \{0,1\}^n$ , check if  $sA \equiv y \pmod{2}$  has a solution (“column reduction”)
- In the pseudorandom case, there will always be a solution!
- But for a uniformly random  $r$ ,  $sA \equiv r \pmod{2}$  is solvable w.p. at most  $2^{\lambda-n}$

**Takeaway:** need some non-linearity in your PRG

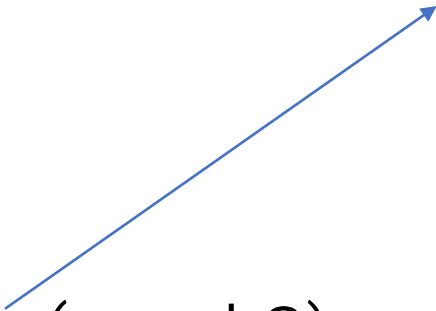
# The LPN Generator

What if we take the subset sum generator **but flip some of the output bits?**

$G(A, s, j_1, \dots, j_\lambda)$ :

- Compute  $y = sA \pmod{2}$
- For all  $j$ ,
  - If  $j = j_i$  for some  $i$ , set  $r_j = y_j \oplus 1$
  - Else set  $r_j = y_j$
  - Output  $(A, r)$

“Random sparse error”

$$r = sA + e \pmod{2}$$


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Output length:  $\lambda \cdot n + n$

Input length:  $\lambda \cdot n + \lambda + \lambda \cdot \log(n)$

Set  $n = \lambda^2$  and we have stretch!

$$r = sA + e \pmod{2}$$

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  - Else set  $r_j = y_j$
  - Output  $(A, r)$
- Attack from earlier doesn't work!
  - $\lambda$  of the equations are wrong, and we don't know which!
- Best known attack:  $2^{\lambda / \log(\lambda)}$  time
- Thm (non-trivial): Breaking  $G$  is as hard as solving for  $s$ !

$$r = sA + e \pmod{2}$$

# The MQ Generator

What if we take subset sums but add in quadratic terms?

$$G(A, B_1, \dots, B_n, s^\top) = (A, B_1, \dots, B_n, s^\top A + (s^\top B_1 s, s^\top B_2 s, \dots, s^\top B_n s))$$

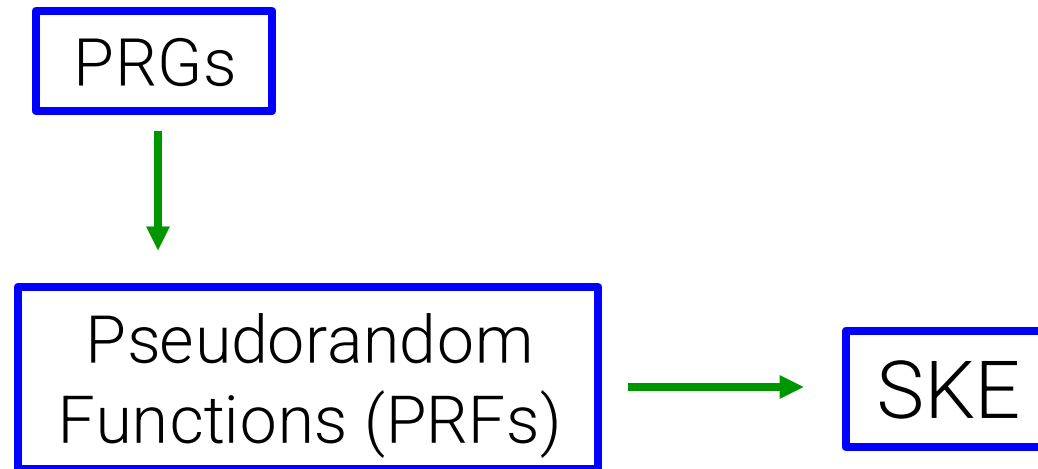
- General quadratic equations modulo 2 are NP-hard to solve!
- ...but we don't know as much about random quadratic equations
- Unlike LPN, we must restrict  $n < \lambda^2$  or else there is an attack! (HW1)

# “The PRG Conjecture”

**Main Conjecture:** PRGs with one bit of stretch exist

If true, we have secret-key cryptography!

# Plan: Secret-key encryption from PRGs



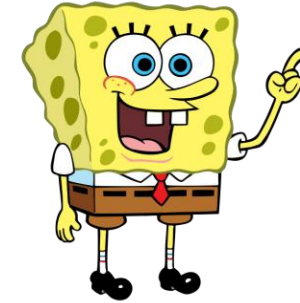


# Secret-Key Encryption (multiple messages)



$k, m$

$$ct = \text{Enc}(k, m)$$



$k$

$$m' = \text{Dec}(k, ct)$$

**Notation:** If  $(\text{Enc}, \text{Dec})$  is a secret-key encryption scheme with message length  $n = n(\lambda)$  and  $\mathbf{m} \in (\{0,1\}^n)^t$ , we define  $\text{Enc}(k, \mathbf{m}) = (\text{Enc}(k, m_1), \dots, \text{Enc}(k, m_t))$

**Definition:** A secret-key encryption scheme with message length  $n = n(\lambda)$  satisfies  $t = t(\lambda)$ -message indistinguishability (IND-) security if:

For all  $\mathbf{m}_0 = \left( \mathbf{m}_0^{(\lambda)} \in (\{0,1\}^{n(\lambda)})^{t(\lambda)} \right)_\lambda, \mathbf{m}_1 = \left( \mathbf{m}_1^{(\lambda)} \right)_\lambda$ ,  $\text{Enc}(\mathbf{k}, \mathbf{m}_0) \approx_c \text{Enc}(\mathbf{k}, \mathbf{m}_1)$ .

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**Observation:** for  $t \geq 2$ , encryption can no longer be deterministic!

$$\text{Enc}(\textcolor{red}{k}, m_0), \text{Enc}(\textcolor{red}{k}, m_0) \approx_c \text{Enc}(\textcolor{red}{k}, m_0), \text{Enc}(\textcolor{red}{k}, m_1)$$

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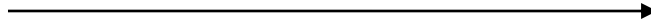
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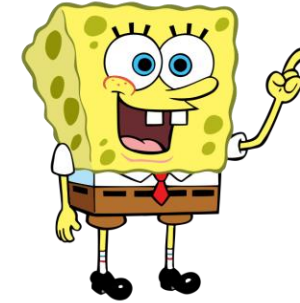
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**Observation:** without loss of generality, we can take  $n(\lambda) = 1$  (one-bit messages)!

Redefine  $\text{Enc}(k, m \in \{0,1\}^n) = (\text{Enc}(k, m_1), \dots, \text{Enc}(k, m_n))$

**Definition:** A secret-key encryption scheme with message length  $n = n(\lambda)$  satisfies **indistinguishability (IND-) security** if **for all**  $t(\lambda) = \text{poly}(\lambda)$ ,

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**Observation:** without loss of generality, we can take  $n(\lambda) = 1$  (one-bit messages)!

**Exercise:**  $nt$ -message security (one-bit messages)  $\rightarrow$   $t$ -message security ( $n$ -bit messages)

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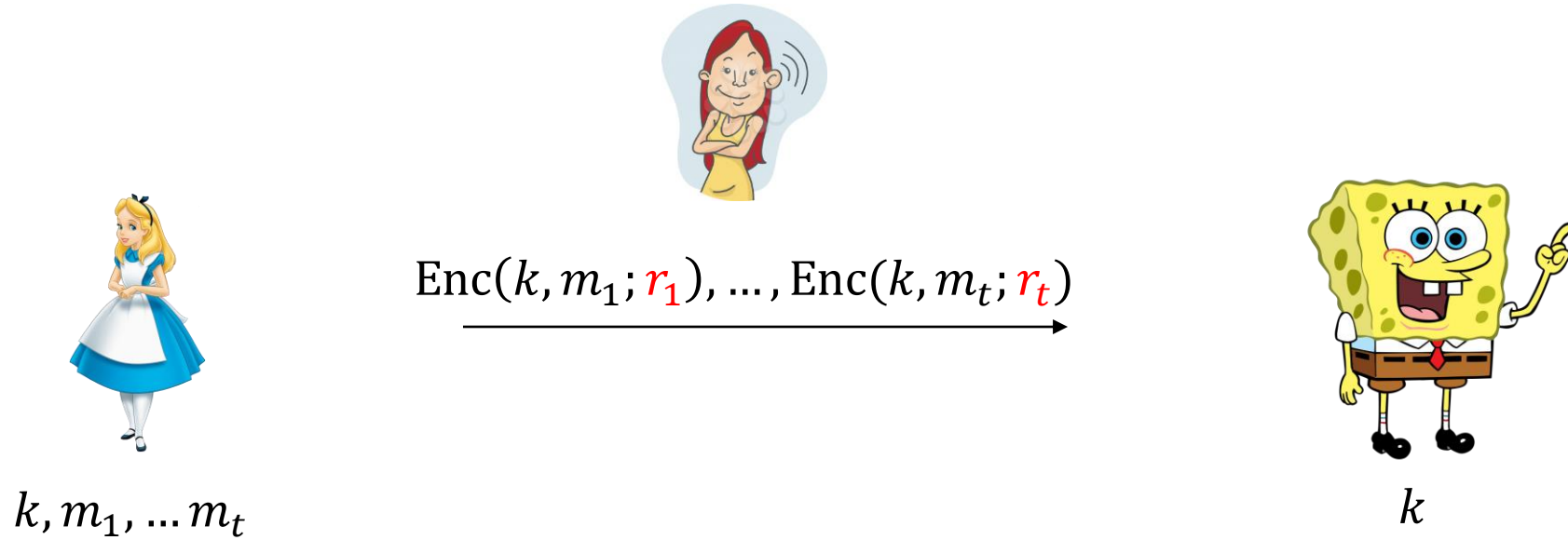
# Secret-Key Encryption (multiple messages)

Is this what we want?

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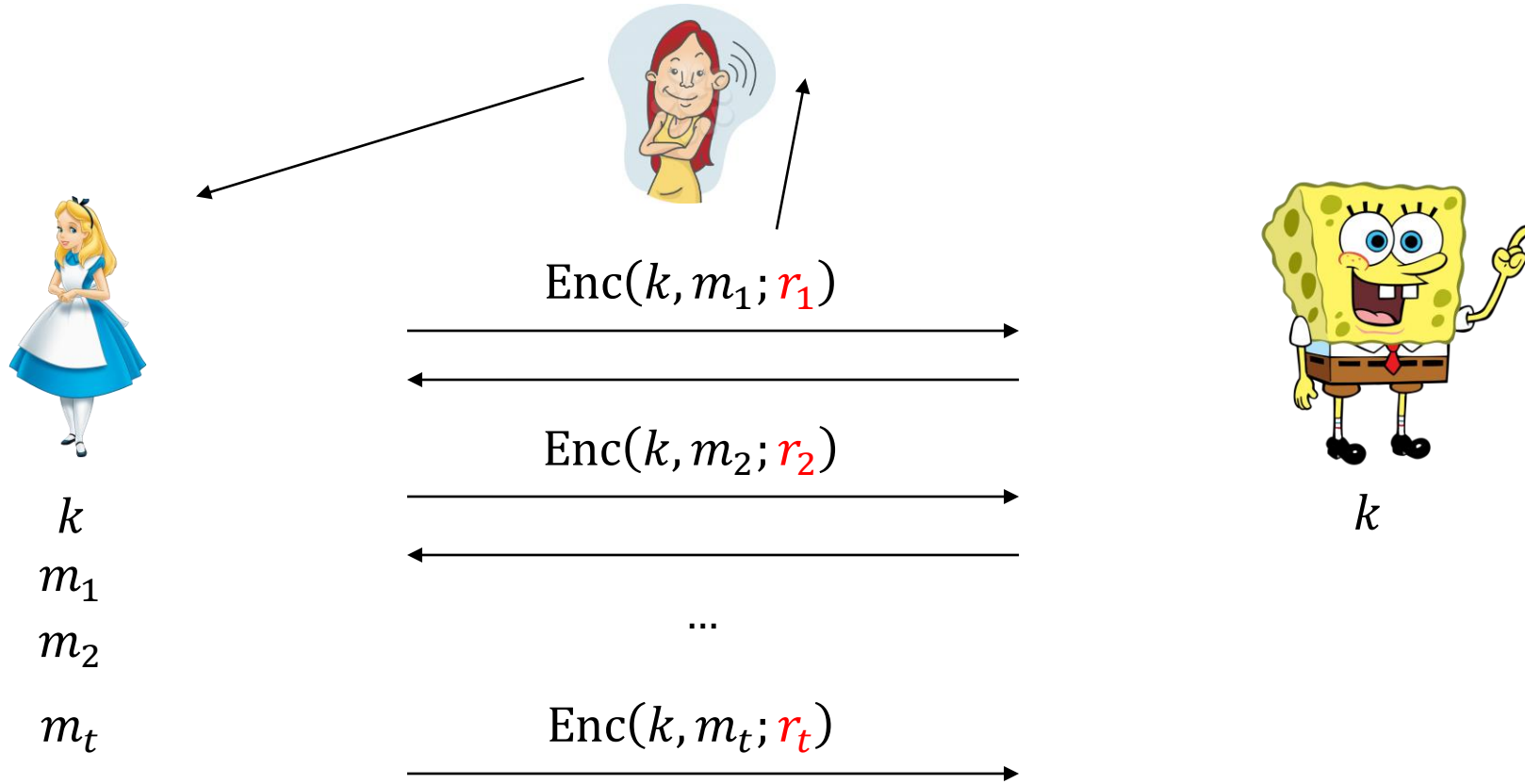
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# Secret-Key Encryption (multiple messages)

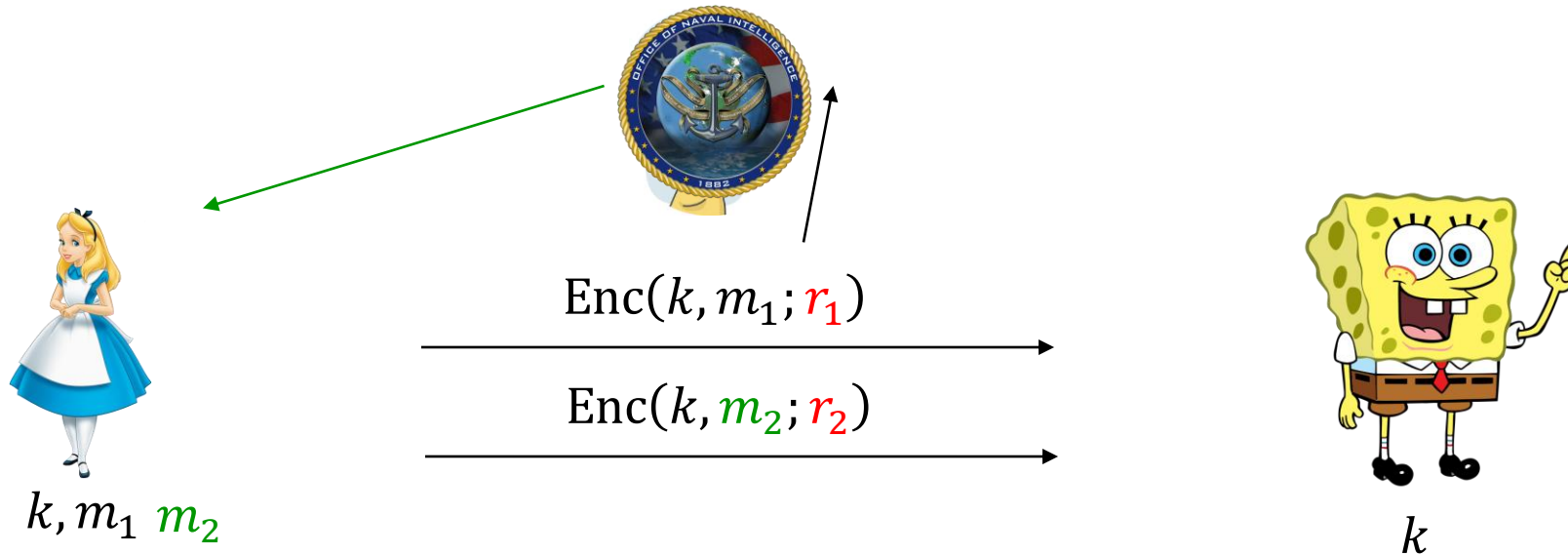


What information **can  $m_i$  depend on?**

- Messages (encrypted or unencrypted) that Bob sends back to Alice
- The weather on day  $i$
- Actions that the adversary  takes.

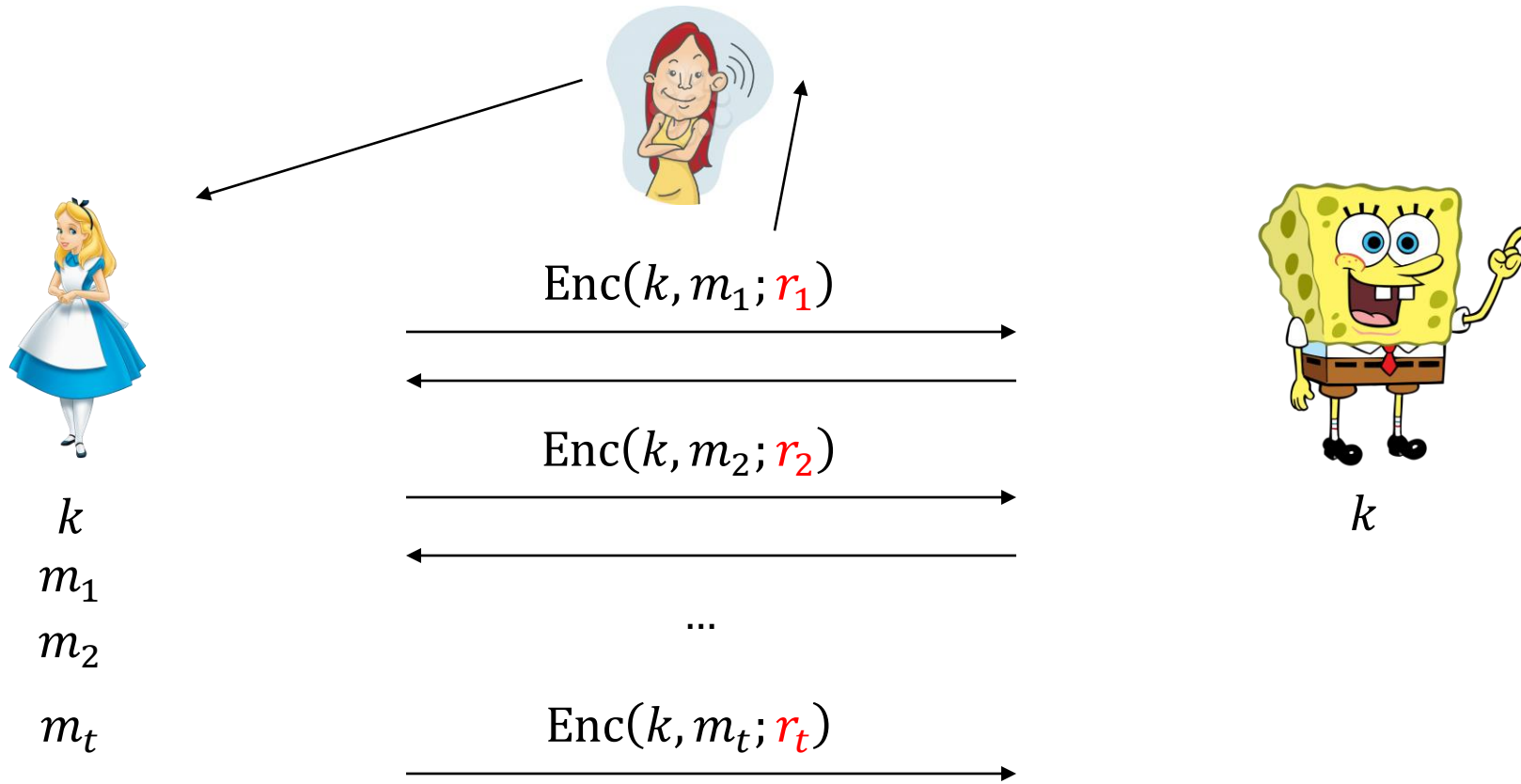


# An “known plaintext attack”



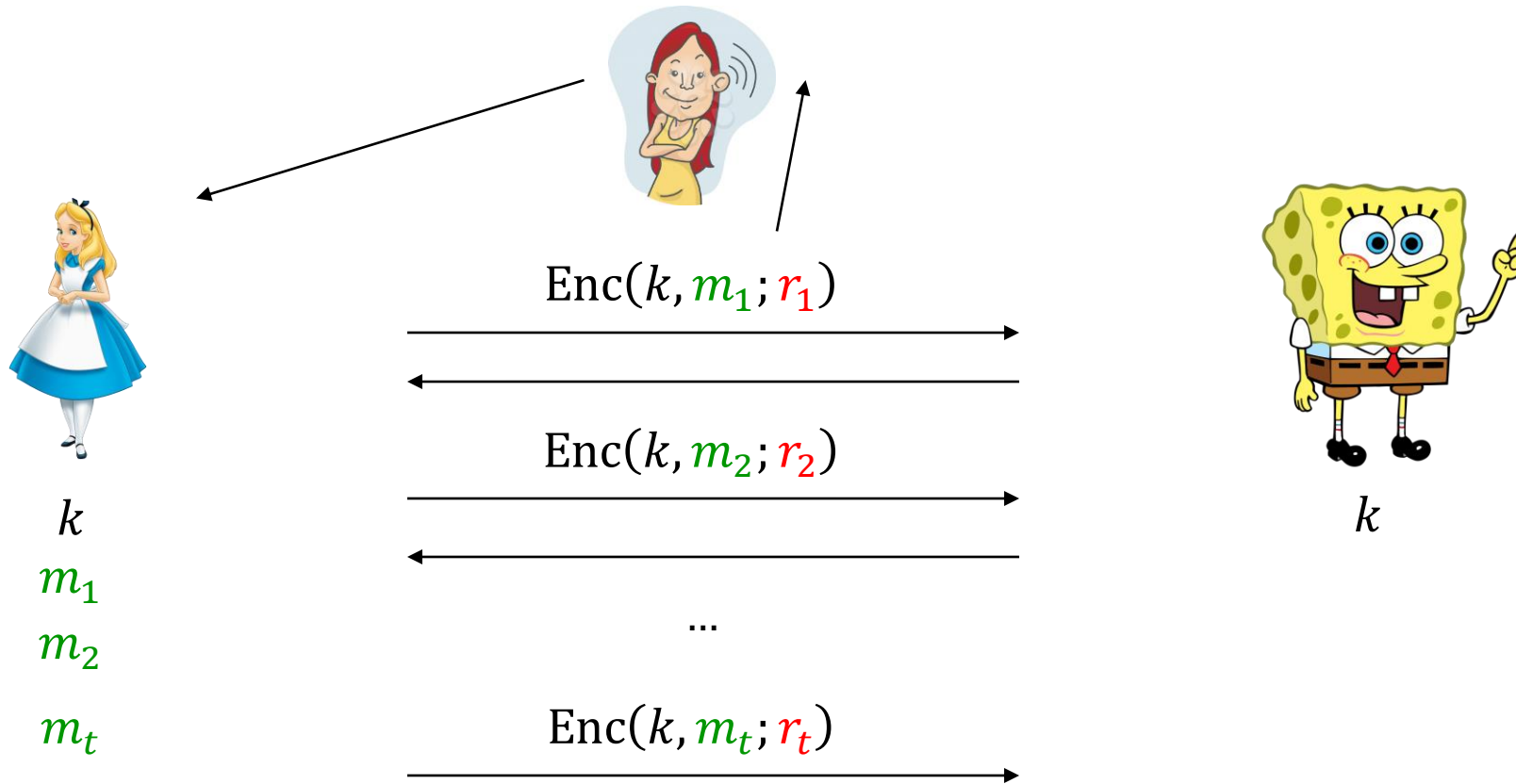
- WWII: US Navy eavesdrops that Japan plans to attack location “AF”
- Previous knowledge: locations near Hawaii have codewords starting with A
- Educated guess: “AF” refers to Midway island
- US releases a fake message about Midway island in the clear (“low supplies”)
- Japan sends reports about “AF” having low supplies.

# Secret-Key Encryption (multiple messages)



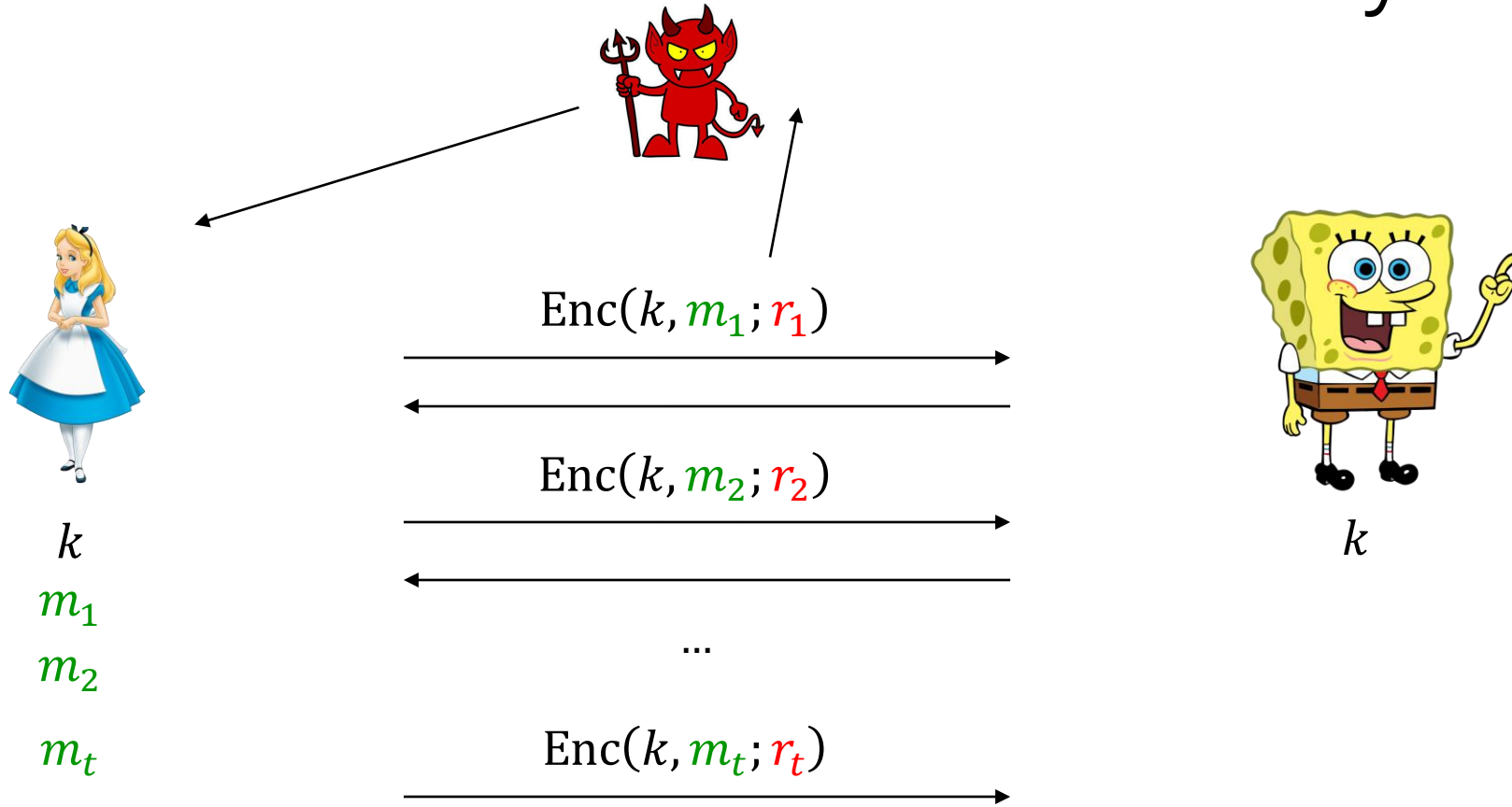
How do we model this dependency?

# Chosen-Plaintext Security



Cautious model: adversary has **complete power** over the choice of messages.

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# Game-based security definitions

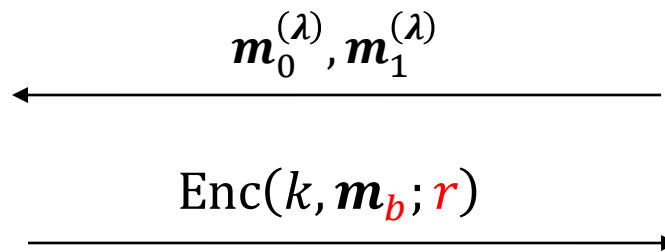
# IND-Security, Again

Challenger/Honest Party



$k$

$b \leftarrow \{0,1\}$



Adversary



$b'$

Win if  $b' = b$

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$m_0^{(\lambda)}, m_1^{(\lambda)}$



$\text{Enc}(k, m_b; r)$



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For all polynomial-time



$$\Pr[\text{Win}] \leq \frac{1}{2} + \text{negl}(\lambda)$$

# IND-CPA Security

Challenger/Honest Party



Sampled once

$k$

$b \leftarrow \{0,1\}$

$m_0^{(\lambda)}, m_1^{(\lambda)}$

$\text{Enc}(k, m_b; r)$

New Enc randomness  
each time

Adversary



$b'$

Win if  $b' = b$

**Definition:** A secret-key encryption scheme with message length  $n = n(\lambda)$  satisfies **indistinguishability (IND-CPA) security** if **for all**  $t(\lambda) = \text{poly}(\lambda)$ ,

For all polynomial-time



$$\Pr[\text{Win}] \leq \frac{1}{2} + \text{negl}(\lambda)$$



# IND-CPA Security, Alt. Definition

Challenger/Honest Party



$k$

$b \leftarrow \{0,1\}$

$m^{(\lambda)}$

$\text{Enc}(k, m; r)$

$m_0^{(\lambda)}, m_1^{(\lambda)}$

Challenge:  $\text{Enc}(k, m_b; r)$

$m^{(\lambda)}$

$\text{Enc}(k, m; r)$

Adversary



$b'$

Win if  $b' = b$

For all polynomial-time



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# IND-CPA Security

**Lemma:** Definition and Alt. Definition are equivalent

**Proof:**

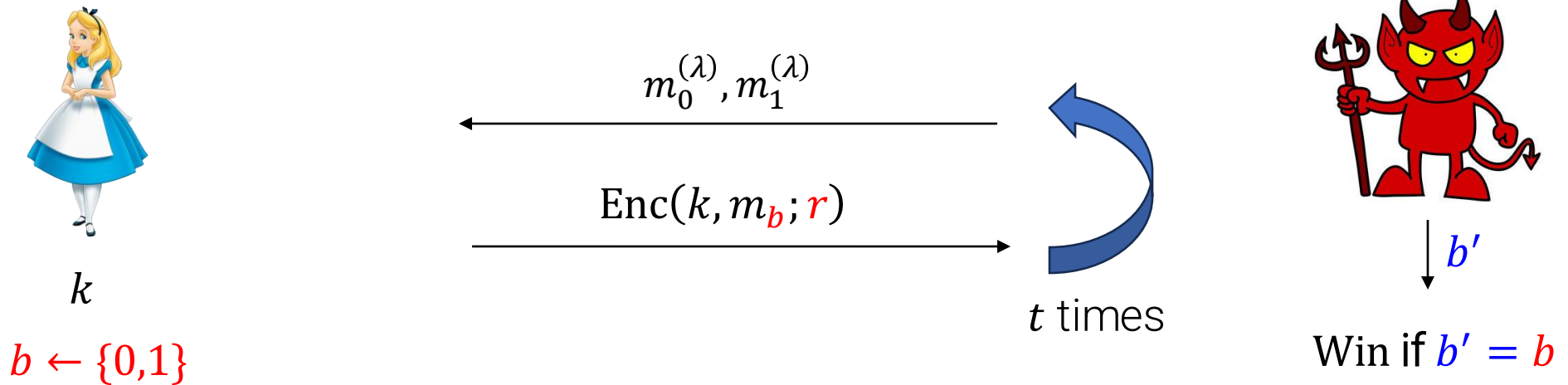
→ Alt. Definition is a special case of Definition

← Hybrid argument & security reduction!

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**Lemma:** Definition and Alt. Definition are equivalent

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Want to show:  $\Pr[\text{Win}] \leq \frac{1}{2} + \text{negl}(\lambda)$

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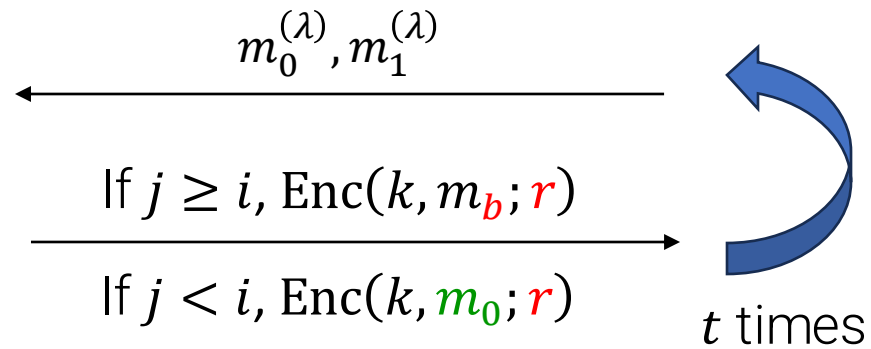
**Proof:** ← Hybrid argument & security reduction!

Game<sub>*i*</sub>



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Want to show:  $\Pr_0[\text{Win}] \leq \frac{1}{2} + \text{negl}(\lambda)$

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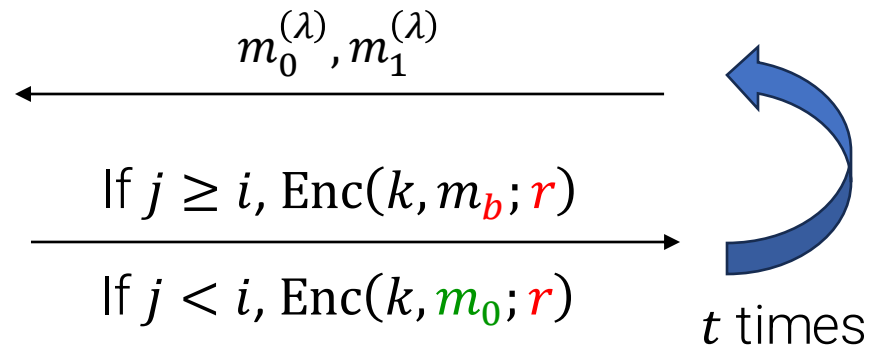
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Observe:  $\Pr_{t+1}[\text{Win}] = \frac{1}{2}$

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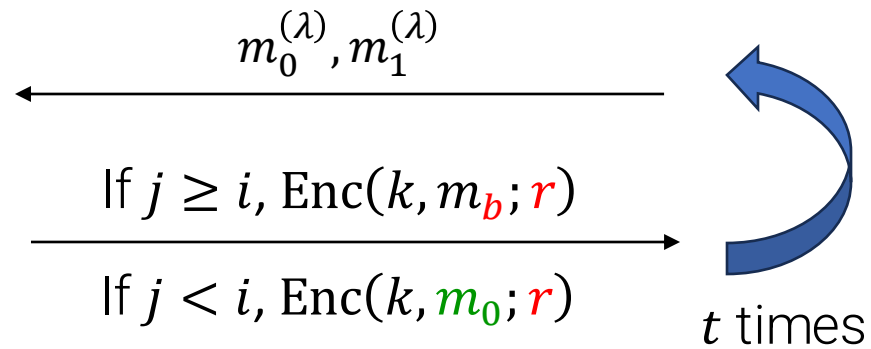
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$$\text{Claim: } \left| \Pr_{i+1}[\text{Win}] - \Pr_i[\text{Win}] \right| = \text{negl}(\lambda)$$

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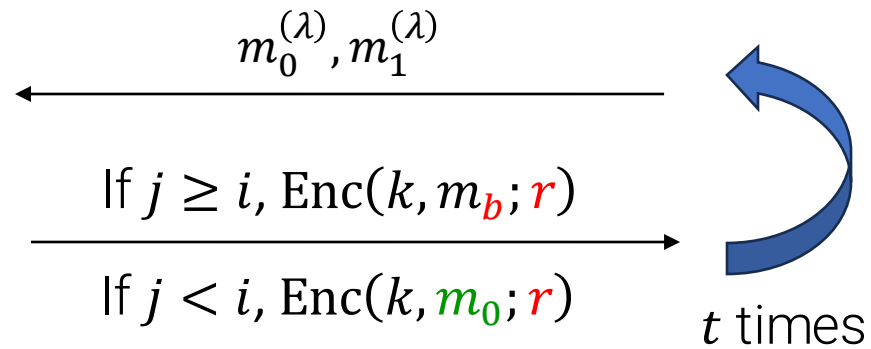
Proof by contradiction: suppose  exists such that  $\left| \Pr_{i+1}[\text{Win}] - \Pr_i[\text{Win}] \right| > \epsilon(\lambda)$

Game<sub>*i*</sub>



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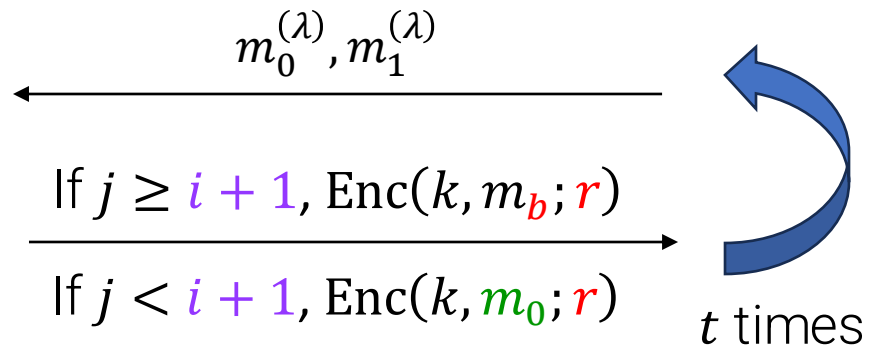
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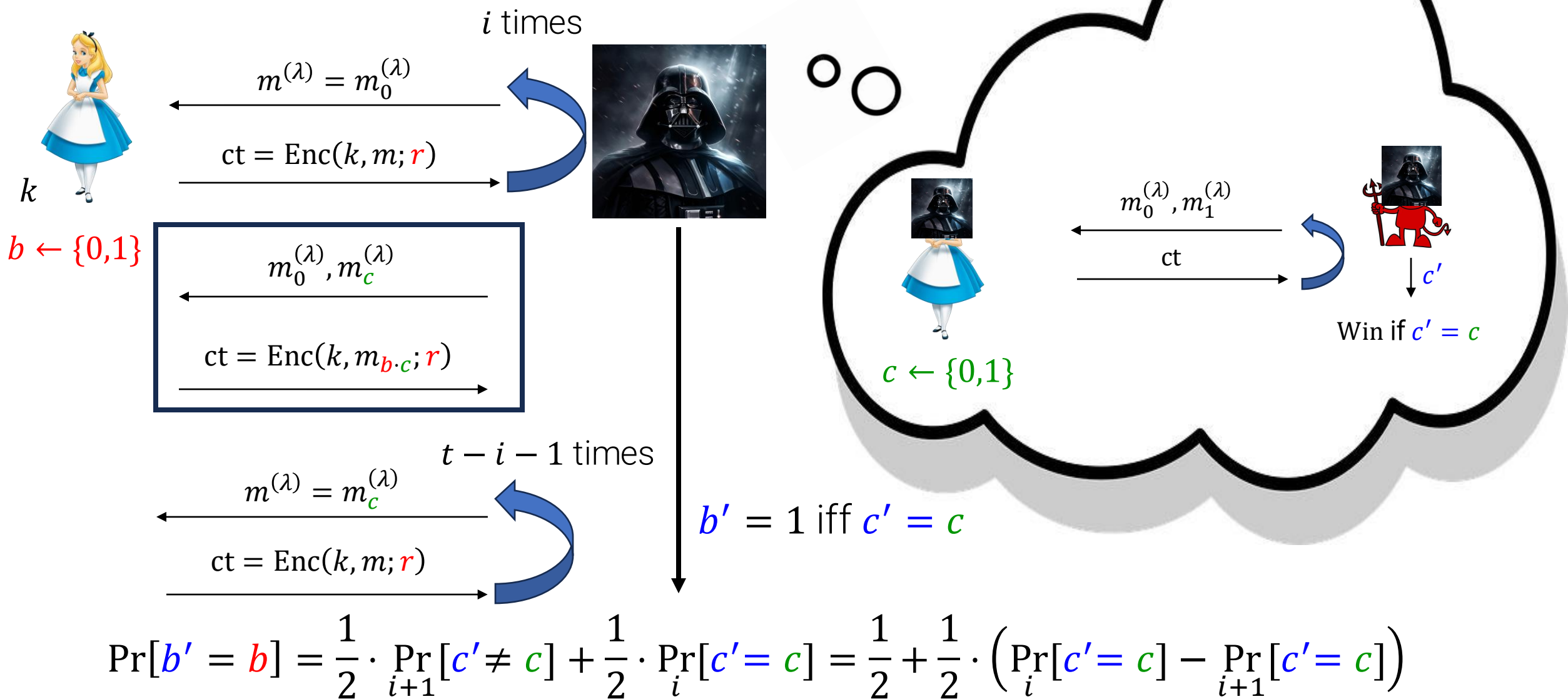
We will construct



violating the alt. definition!



# IND-CPA Security



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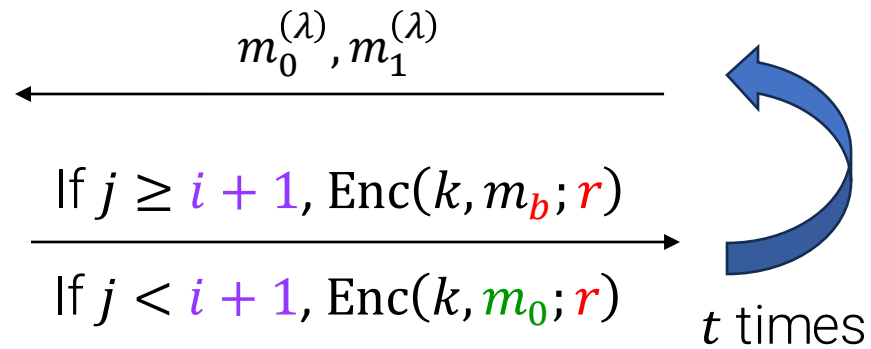
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